THE ACTUARIAL SOCIETY OF KENYA

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THE NEXT FRONTIER: ACTUARIES BEYOND BOUNDARIES

ARDHIGUARD

PREDICT. PROTECT. PRESERVE LAND.

AN AI-ACTUARIAL FRAMEWORK FOR PREDICTING AND PREVENTING LAND LOSS THROUGH LOAN DEFAULT IN KENYA

BY

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SUBMITTED TO THE ACTUARIAL SOCIETY OF KENYA

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DECLARATION

I declare that this paper is my own original work and has not been presented for any other award. Where the work of others has been used it has been acknowledged and referenced in accordance with academic convention.

The empirical analysis reported in Sections 5 and 6 was conducted by the author in R. The dataset used is publicly available and its provenance is stated in full in Section 5.1.

Ndegwa Wangui

Sign: ---------------------------------- Date: --

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ABBREVIATIONS AND NOTATION

API: Application Programming Interface

CAF: Climate Adjustment Factor

CBK: Central Bank of Kenya

DTI: Debt-to-Income ratio

EAD: Exposure at Default

EL: Expected Loss

FSD: Financial Sector Deepening

GP: Gross Premium

IRA: Insurance Regulatory Authority of Kenya

KES: Kenya Shilling

KMD: Kenya Meteorological Department

KNBS: Kenya National Bureau of Statistics

LGD: Loss Given Default

LTV: Loan-to-Value ratio

MFI: Microfinance Institution

MSE: Micro and Small Enterprise

MU: Monetary Unit, the unit of the source dataset; see Section 5.1

NPL: Non-Performing Loan

SACCO: Savings and Credit Cooperative Organisation

SPI: Standardised Precipitation Index

TASK: The Actuarial Society of Kenya

WMO: World Meteorological Organisation

Notation used throughout the paper is as follows.

Symbol	Meaning	Value or source
q	Probability of default over one policy period	0.11613, estimated, Section 6.1.1
q_i	Default rate for rating class i	Table 3
L	Exposure at default, average loan amount	127,578.9 MU, Section 6.1.2
r	Recovery rate	0.40, assumed, Section 6.1.3
θ	Loading factor	0.20, assumed, Section 6.1.4
EL	Expected loss per insured borrower	8,889.31 MU, Equation 4
GP	Base annual gross premium	10,667.17 MU, Equation 5
C_i	Credibility rating factor for class i	q_i / q, Equation 10
P_i	Risk-tiered premium for class i	$GP \times C_i$, Equation 9
n	Number of insured borrowers in the pool	Varies
N	Number of default claims in one policy period	Random variable
λ	Poisson claim intensity	$n \times q$
α	Negative binomial overdispersion parameter	0.01, illustrative, Section 6.3.1
β	Climate sensitivity coefficient	0.20, assumed, Section 6.5.4
CAF	Climate Adjustment Factor	Equation 11
N_{99}	99th percentile of the claim count distribution	Table 5

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ABSTRACT

Kenya's gross non-performing loan ratio reached 16.1 per cent in April 2024, representing KES 651.8 billion in impaired credit. Mortgage and land-secured loan defaults reached a record KES 40 billion in 2023, and micro and small enterprise default rates surged to 60.7 per cent within a single quarter. Behind each of these statistics is a borrower who pledged land as collateral and now faces losing it permanently. Land foreclosure in Kenya removes a household's primary productive asset, disrupts food production, interrupts schooling and eliminates generational wealth in a single court order.

This paper introduces ArdhiGuard, an AI-actuarial framework designed to predict loan default risk and intervene before Kenyan families lose land-secured collateral. ArdhiGuard integrates three interdependent layers: a risk engine that monitors six weighted borrower risk factors to generate early warnings ahead of default; a community insurance pool through which borrowers contribute actuarially priced premiums into a shared protection fund; and a mobile money integrated platform for digital premium collection, automated alerts and real-time intervention.

The actuarial model is grounded in an anonymised dataset of 255,347 borrower records, from which a default probability of $q = 0.11613$ and an average exposure of $L = 127{,}578.9$ monetary units are derived. Using an expected loss framework with a 40 per cent recovery rate and a 20 per cent loading factor, the base annual gross premium is 10,667.17 monetary units. Risk-tiered credibility factors differentiate premiums by observed credit quality, from 9,342.31 for excellent-credit borrowers to 11,464.01 for poor-credit borrowers, a spread of 22.7 per cent. A dynamic Climate Adjustment Factor links the Standardised Precipitation Index to premium levels in real time and constitutes the principal methodological contribution of the framework.

The paper also reports a structural finding that follows directly from the model. Under the stated loading and a 150 per cent solvency margin, single-year premium income converges to approximately 80 per cent of the required reserve as the pool grows, so the reserve requirement cannot be met from premium income alone at any pool size. Seed capital and quota share reinsurance are therefore structural requirements of the design rather than optional refinements.

ArdhiGuard demonstrates that actuarial science, combined with predictive analytics and mobile money infrastructure, can move beyond retrospective risk pricing into proactive social protection. It contributes an implementable model for land-secured loan insurance in emerging markets, with direct alignment to Kenya's financial inclusion agenda.

Keywords: loan default prediction; land foreclosure; actuarial risk pricing; microinsurance; early warning systems; climate risk; inclusive finance; Kenya; non-performing loans; collateral protection; Standardised Precipitation Index.

JEL Classification: G22, G28, Q54, O16.

1. INTRODUCTION

Land is the most consequential asset a Kenyan family can own. It provides food, shelter, collateral for credit and the foundation of intergenerational wealth transfer. Yet this asset, irreplaceable by nature, is increasingly at risk. As Kenya's formal credit market has expanded to reach smallholder farmers, micro-enterprises and informal sector workers, the number of borrowers using land as loan collateral has grown substantially. So too has the number of those who, when facing income shocks, find themselves unable to repay.

The consequences are not merely financial. When a bank forecloses on land-secured collateral, families lose their primary productive asset at auction prices that have historically been discounted by up to 30 per cent below market value. The borrower receives nothing. The land is gone. The family is displaced. Children leave school. Agricultural output falls. Rural poverty deepens.

This paper argues that this outcome, which current financial infrastructure treats as inevitable, is in fact predictable and preventable. The tools to prevent it already exist. Actuarial risk science can price and pool the risk. Predictive analytics can detect early warning signals before default crystallises. Mobile money infrastructure already deployed across Kenya can deliver protection at scale. What has been missing is a framework that combines all three.

ArdhiGuard is that framework. It is an AI-actuarial system designed to identify borrowers at risk of default in advance of the event, trigger an insurance-funded intervention, and preserve land ownership while simultaneously protecting the lender's loan book. This paper presents the theoretical and actuarial basis for ArdhiGuard, demonstrates its financial characteristics through empirically grounded modelling, and proposes an implementation roadmap grounded in Kenya's existing regulatory and technological infrastructure.

1.1 Scope and objectives

This paper addresses four objectives:

(i) To quantify the scale and severity of land-secured loan default in Kenya using current Central Bank of Kenya data.

(ii) To present an actuarially rigorous premium pricing model incorporating expected loss methodology, a Poisson claim frequency distribution with a negative binomial extension, and risk-tiered credibility scoring, grounded in an empirical dataset of 255,347 borrower records.

(iii) To demonstrate the integration of predictive early warning with insurance pool mechanics through a dynamic, climate-responsive pricing framework.

(iv) To establish ArdhiGuard as a replicable model for land-secured loan protection across emerging markets in sub-Saharan Africa.

1.2 Structure of the paper

Section 2 quantifies the problem. Section 3 situates the framework within the existing literature on agricultural microinsurance, credit risk modelling and climate-linked default. Section 4 sets out the three-layer architecture. Section 5 describes the data. Section 6 develops the actuarial methodology and constitutes the analytical core of the paper. Section 7 examines financial viability and reports a structural solvency finding. Sections 8 and 9 address implementation and social impact. Section 10 states the limitations that define the research agenda, and Section 11 concludes.

2. THE PROBLEM: LAND LOSS AS A FINANCIAL AND SOCIAL CRISIS

2.1 Kenya's non-performing loan crisis

Kenya's banking sector is experiencing its worst non-performing loan performance in nearly two decades. The Central Bank of Kenya Bank Supervision Annual Report 2023 recorded a gross NPL ratio of 14.8 per cent at year-end 2023 (Central Bank of Kenya, 2023a), rising to 16.1 per cent by April 2024 and representing KES 651.8 billion in non-performing credit (Central Bank of Kenya, 2024). This deterioration reflects multiple compounding pressures: elevated inflation, rising interest rates following global monetary tightening, drought-related agricultural income shocks and persistent post-pandemic recovery challenges in the micro and small enterprise sector.

Mortgage-specific defaults are particularly alarming. Land and mortgage loan defaults reached KES 40 billion in December 2023, a 14.4 per cent default rate on that portfolio and a record high. Every percentage point increase in this ratio represents hundreds of Kenyan families entering the foreclosure pipeline.

2.2 The micro and small enterprise vulnerability

Micro and small enterprises represent the backbone of Kenya's informal economy, employing an estimated 14.9 million Kenyans. Yet this sector is acutely vulnerable to credit stress. The CBK MSE Tracker Survey of June 2023 found that MSE loan default rates surged from 42.8 per cent to 60.7 per cent within a single quarter, driven by deteriorating business conditions, reduced consumer spending and the cumulative burden of high-interest digital credit (Central Bank of Kenya, 2023b). The majority of MSE borrowers who use formal credit channels secure their loans against land.

2.3 The agricultural exposure gap

Agriculture employs 40 per cent of Kenya's total population and 70 per cent of the rural workforce, contributing 21.8 per cent of GDP (Kenya National Bureau of Statistics, 2023). Yet of Kenya's total KES 4.1 trillion commercial bank loan portfolio in 2023, only approximately 3 per cent reached the agricultural sector. This represents a fundamental market failure: the sector carrying the greatest climate-related default risk receives the least financial protection infrastructure. When agricultural loans default, the land that secured them is lost.

2.4 The foreclosure mechanism and its consequences

Under the Land Act 2012 and the Insolvency Act 2015, a chargee bank has the statutory right to sell charged land upon borrower default following required notice periods (Republic of Kenya, 2012). Pre-reform practice saw banks auctioning defaulted properties at discounts of up to 30 per cent below market value. While the Land Laws (Amendment) Act 2016 introduced protective provisions, enforcement remains inconsistent, particularly in rural counties where land registries are under-digitised and legal representation is inaccessible (Republic of Kenya, 2016).

The consequences extend beyond the individual borrower. Families displaced from agricultural land lose their primary income source, food production capacity and cultural identity. Children are removed from school. Extended family networks are destabilised. The community loses a productive agricultural unit. These second-order effects compound into regional poverty that is difficult to reverse across generations.

3. LITERATURE REVIEW

3.1 Agricultural and land-secured microinsurance

Index-based agricultural insurance has been explored extensively in sub-Saharan Africa as a mechanism for protecting smallholder farmers against weather-related income shocks (Barnett and Mahul, 2007; Jensen, Barrett and Mude, 2016). Kenya's Agriculture Insurance Fund, established under the Agriculture (Amendment) Act 2018, provides area yield and weather index products. However, these instruments protect crop income and not the land itself, and they do not address the loan default mechanism that triggers foreclosure. The gap between agricultural insurance and land asset protection remains substantially unaddressed in the literature.

3.2 Credit risk scoring and predictive analytics in emerging markets

Machine learning applications in credit risk have demonstrated strong predictive performance across emerging market contexts. Khandani, Kim and Lo (2010) showed that consumer transaction data improves default prediction significantly over traditional credit bureau scores. In Kenya specifically, the digitisation of mobile money transaction data has enabled alternative credit scoring by a number of digital lenders. However, these models are designed for loan origination decisions and not for ongoing monitoring to trigger intervention before default.

The concept of early warning systems for loan default has parallels in the corporate distress prediction literature, from Altman's Z-score (Altman, 1968) to more recent ensemble machine learning approaches (Lessmann et al., 2015). ArdhiGuard adapts this tradition to the retail land-secured borrower context through a six-factor credibility-weighted scoring model.

3.3 Insurance pool mechanics and microinsurance design

The actuarial literature on microinsurance design emphasises the tension between affordability and actuarial sustainability (Churchill and Matul, 2012). Premium loading factors must be calibrated carefully. Too high, and low-income borrowers cannot afford coverage. Too low, and the pool becomes insolvent under adverse claim scenarios. Carter et al. (2017), reassessing two decades of index insurance experimentation, find that take-up has remained disappointingly low without sustained subsidy, and that basis risk, the mismatch between the insured index and actual losses, remains a central design problem. Jensen, Mude and Barrett

(2018) document how basis risk and spatiotemporal adverse selection jointly depress demand in northern Kenya.

ArdhiGuard addresses this through individual-level risk monitoring rather than index proxies, which substantially reduces basis risk at the individual borrower level. The climate index in this framework adjusts the price of cover rather than triggering the payout, so an SPI reading that does not match a given borrower's experience changes what they pay, not whether they are protected.

3.4 Climate risk and loan default

The relationship between climate shocks and loan default in agricultural settings is well established. Jensen, Barrett and Mude (2016) document the relationship between drought severity and livestock mortality in northern Kenya, with direct implications for loan repayment capacity. The Standardised Precipitation Index, originally developed by McKee, Doesken and Kleist (1993) and standardised by the World Meteorological Organisation (2012), has been validated as a leading indicator of agricultural income stress across sub-Saharan Africa. Its integration into actuarial pricing models represents a natural extension of climate risk science to insurance applications.

3.5 The regulatory context

Kenya's Insurance Regulatory Authority Microinsurance Regulations 2017 provide a framework for low-premium, simplified insurance products targeting low-income populations (Insurance Regulatory Authority, 2017). The regulations permit group-based premium collection, simplified underwriting and mobile money settlement, all of which are foundational to ArdhiGuard's delivery model. The IRA 2023 Insurance Industry Annual Report notes that microinsurance penetration in Kenya remains below 2 per cent of the insurable population (Insurance Regulatory Authority, 2023), and the FinAccess Household Survey confirms that formal insurance uptake continues to lag well behind formal savings and credit access (FSD Kenya, 2024). Together these confirm the scale of the market opportunity.

4. THE ARDHIGUARD FRAMEWORK

ArdhiGuard operates as a three-layer system in which each component is necessary and interdependent. The analytics layer generates the intelligence. The insurance layer provides the financial mechanism. The delivery layer carries both at scale.

4.1 Layer one: the risk engine

The risk engine is the predictive core of ArdhiGuard. It monitors six borrower risk factors and produces a composite risk score that is updated monthly or triggered by significant data events. The six factors and their indicative weights are presented in Table 1.

Table 1: Risk engine, six-factor weighting framework

Risk factor	Weight	Rationale
Payment history, recency and pattern	25%	Strongest individual predictor of future default
Income stability	20%	Irregular income is the primary driver of cash-flow default
Debt-to-income ratio	20%	Measures existing burden relative to repayment capacity
Weather and climate exposure (SPI)	15%	Drought and flood events directly reduce agricultural income
Loan-to-value ratio	12%	High LTV increases loss severity upon default
Business sector risk	8%	Sector-level stress amplifies individual default probability

4.1.1 Status of the weighting scheme

The weights in Table 1 are expert judgement, not estimated parameters. They reflect the ordering of predictive importance reported in the credit scoring literature, but they have not been fitted to the dataset used in Sections 5 and 6, and the paper does not claim otherwise. The same applies to the intervention threshold and the lead time discussed below.

A note on what this layer is and is not. *Layer one as specified in this paper is a transparent, fixed-weight scoring rubric. It is not a trained model. The intended estimation route is a penalised logistic regression fitted to the borrower records described in Section 5, with the six weights derived from standardised coefficients and validated against a held-out sample, followed by comparison with a gradient boosted* *alternative to test whether the loss of interpretability buys enough predictive lift to justify it. Interpretability matters here: a borrower who receives an intervention alert is entitled to know which factor moved, and a fixed-weight rubric answers that question in a way that an ensemble does not. Until that estimation is completed, the weights should be read as a specification of what the engine measures rather than a claim about how well it predicts. This is recorded as a limitation in Section 10.*

When the composite score crosses a predefined risk threshold, provisionally set at a 15 per cent default probability, the system triggers an intervention alert. The borrower receives an SMS notification. The lending institution is notified. An intervention officer initiates contact. The design target is a lead time of 60 to 90 days ahead of the expected default date, which is the window within which a restructuring or bridge payment can realistically be arranged. Both the threshold and the achieved lead time are quantities to be measured in the pilot, not results of the present model.

4.2 Layer two: the insurance pool

The insurance pool is the financial engine that makes intervention possible. It operates on the principle of risk pooling. Many borrowers contribute premiums so that the few who face crisis receive meaningful support. Premium levels are actuarially derived in Section 6 and risk-tiered so that higher-risk borrowers contribute proportionally more.

Upon triggering, the fund can deploy three forms of intervention: temporary loan instalment coverage, providing bridge payments directly to the lending institution; negotiation support, engaging the bank to restructure the loan; and emergency income support, providing targeted grants to address an immediate income shock. The appropriate intervention type is determined by the nature of the triggering event.

4.3 Layer three: the delivery platform

ArdhiGuard is delivered through a mobile-first platform integrated with mobile money, USSD codes and a smartphone application. Premium collection is automated through standing mobile money instructions, which eliminates collection friction, the single largest cost driver in low-premium distribution. Risk score updates are pushed to borrowers by SMS, giving them visibility of their own financial position. For lending institutions, an API provides real-time risk dashboards across their land-secured loan portfolio.

5. DATA

5.1 Data source and its limits

The actuarial pricing model was developed using a publicly available anonymised loan default dataset containing borrower-level variables including loan amount, income, employment type, credit score, debt-to-income ratio, loan purpose and default status. The dataset is drawn from an international lending context and is denominated in unspecified monetary units rather than Kenya Shillings.

Currency convention used throughout this paper. *Every monetary quantity derived from the dataset is expressed in monetary units, abbreviated MU. KES is used only for genuinely Kenyan quantities drawn from published sources, such as the Central Bank of Kenya statistics in Section 2 and the seed capital estimate in Section 7. Readers should treat all MU premium figures as methodologically illustrative. Kenyan implementation requires recalibration on loan-level data in KES from a domestic microfinance institution or SACCO partner, which is identified as the highest-priority next step in Section 10.*

5.2 Data preparation

The dataset was imported into R for preparation and analysis. Its structure, data types and variable names were verified. The dataset contained 255,347 borrower records across 18 variables. Checks for missing values and duplicate records returned no observations requiring removal. Of the 18 variables, 8 were retained for modelling: age, income, loan amount, credit score, interest rate, employment type, loan purpose and default status. The remainder were administrative identifiers with no predictive content.

5.3 Exploratory data analysis

Of the 255,347 borrower records, 225,694 did not default and 29,653 defaulted, corresponding to an empirical default probability of:

$$q = 29{,}653 \,/\, 255{,}347 = 0.11613 \tag{1}$$

The average loan amount across all borrowers was L = 127,578.9 MU. These two statistics provide the primary inputs for the expected loss model. Summary statistics are presented in Table 2.

Table 2: Summary statistics, loan default dataset

Statistic	Value
Total borrowers	255,347
Non-defaults	225,694
Defaults	29,653
Empirical default probability (q)	0.11613
Average loan amount (L)	127,578.9 MU

Table 3 shows default rates stratified by credit score category. A monotonic relationship is evident: borrowers with higher credit scores default at lower rates, which confirms credit score as a valid actuarial rating variable. The 2.31 percentage point spread from Poor to Excellent is sufficient to support meaningful premium differentiation.

Table 3: Default rate and rating factor by credit score category

Credit score category	Non-default	Default rate (q_i)	Rating factor (C_i)
Poor	87.52%	12.48%	1.0747
Fair	88.58%	11.42%	0.9834
Good	89.37%	10.63%	0.9154
Excellent	89.83%	10.17%	0.8758

$Ci = \frac{q_i}{q}, where\ q = 0.11613$ *is the pool average default rate.*

Table 4 shows default rates by loan purpose, ordered from highest to lowest risk. Business loans carry the highest default risk at 12.33 per cent and home loans the lowest at 10.23 per cent. The 2.10 percentage point range is comparable to the credit score range and supports loan purpose as a supplementary rating variable.

Table 4: Default rate by loan purpose

Loan purpose	Non-default	Default rate
Business	87.67%	12.33%
Auto	88.12%	11.88%
Education	88.16%	11.84%
Other	88.21%	11.79%
Home	89.77%	10.23%

6. ACTUARIAL METHODOLOGY

The exploratory analysis in Section 5 provides the empirical basis for estimating the four parameters required for the premium pricing model. Sections 6.1.1 through 6.1.4 derive each in turn.

6.1 Parameter estimation

6.1.1 Probability of default (q)

The probability of default represents the likelihood that a borrower fails to repay during the policy period. It is estimated directly from the dataset as the proportion of borrowers who defaulted:

$$q = \frac{number\ of\ defaults}{total\ borrowers} = \frac{29{,}653}{255{,}347} = 0.11613 \quad (2)$$

The unrounded value is carried through all derived calculations to avoid rounding-induced discrepancies. The policy period is one year, so q is an annual default probability and every premium derived from it is an annual premium. This convention is stated explicitly here because it governs the reserve and revenue calculations in Section 7.

6.1.2 Exposure at default (L)

Exposure at default is approximated by the average loan amount across all borrowers in the dataset:

$$L = \sum \frac{loan\ amount_i}{n} = 127{,}578.9\ MU \quad (3)$$

Using the average loan amount as a uniform exposure is a simplifying assumption appropriate for pool-level pricing. In individual-level implementation, each borrower's outstanding loan balance at the time of risk assessment would be used instead.

6.1.3 Recovery rate (r)

The recovery rate represents the proportion of the outstanding loan expected to be recovered after default through collateral realisation, legal recovery procedures or negotiated settlement. The dataset contains no borrower-level recovery information, so a constant recovery rate of r = 0.40 is assumed. This implies a loss given default of 60 per cent.

On the choice of benchmark, and the direction of its bias. *The 40 per cent figure sits within the range commonly reported in the credit risk literature for partially secured consumer lending, and is broadly consistent with the Basel II standardised treatment of retail exposures (Basel Committee on Banking Supervision, 2006). It is important to be explicit that this is not a like-for-like benchmark. ArdhiGuard insures land-secured exposures, and secured exposures generally recover at higher rates than unsecured ones, which would make r = 0.40 conservative. Pulling in the opposite direction, Kenyan post-foreclosure realisations have historically been depressed by auction discounting of up to 30 per cent below market value, which would make r = 0.40 optimistic. The two biases work against each other and the net direction cannot be signed without domestic data. Calibration against Land Registry auction records and partner bank post-foreclosure recovery data is a priority deliverable for the pilot phase described in Section 8.*

6.1.4 Loading factor (θ)

The loading factor represents the margin added to the pure premium to cover administrative expenses, claim handling costs, contingency reserves and profit. A loading factor of $\theta = 0.20$ is applied. An indicative decomposition is operational expenses covering staff, technology and mobile money transaction fees at 12 per cent, a contingency reserve at 5 per cent, and a profit margin at 3 per cent.

This decomposition is indicative and requires a detailed expense allocation study at implementation. It is carried forward explicitly because the 5 per cent contingency component turns out to drive the solvency result reported in Section 7.2.

6.2 Expected loss and premium calculation

6.2.1 Expected loss

The expected loss per insured borrower over one policy year is:

$$EL = q \times L \times (1 - r) = 0.11613 \times 127{,}578.9 \times 0.60 = 8{,}889.31\,MU \quad (4)$$

6.2.2 Base gross premium

Applying the loading factor $\theta = 0.20$:

$$GP = EL \times (1 + \theta) = 8{,}889.31 \times 1.20 = 10{,}667.17\,MU \quad (5)$$

This is the base annual gross premium for a borrower of average credit quality under normal climatic conditions, before any credibility adjustment or climate repricing. For collection purposes the annual premium is divided into twelve monthly instalments of 888.93 MU, recovered through standing mobile money instructions. The distinction between the annual premium and the monthly instalment is maintained throughout the paper.

6.3 Poisson claim frequency model

The frequency of borrower defaults within an insurance pool is modelled using the Poisson distribution, which is appropriate for rare, independent events occurring within a fixed population over a specified period. Let N denote the number of default claims during one policy year. Then:

$$N \sim Poisson(\lambda),\ \lambda = n \times q \qquad (6)$$

where n is the number of insured borrowers. The probability of observing exactly k defaults is:

$$P(N = k) = e - \lambda \times \lambda k \,/\, k! \qquad (7)$$

Under the Poisson assumption the variance equals the mean, so the standard deviation is the square root of λ. Table 5 reports the resulting distributional quantities for three illustrative pool sizes.

Table 5: Expected claim frequency for illustrative pool sizes (q = 0.11613)

Pool size (n)	$\boldsymbol{\lambda = E[N]}$	**SD(N) =**$\boldsymbol{(r)\lambda}$	**99th percentile**
500	58.07	7.62	76
1,000	116.13	10.78	141
2,000	232.26	15.24	268

99th percentile estimated as lambda + 2.326 squareroot(lambda)using the normal approximation, which is adequate at these values of λ.

6.3.1 Overdispersion and the negative binomial extension

The Poisson model assumes defaults are independent across borrowers. This assumption is reasonable under normal economic conditions. During drought events, however, many borrowers in the same county face the same climate shock simultaneously, which induces positive correlation in defaults and produces a claim count whose variance exceeds its mean. This is precisely the exposure that ArdhiGuard is designed to cover, so the Poisson tail understates the quantity that matters most.

Overdispersion is accommodated by replacing the Poisson with a negative binomial, under which:

$$Var[N] = \lambda(1 + \alpha\lambda) \tag{8}$$

for an overdispersion parameter $\alpha > 0$, with the Poisson recovered in the limit as α approaches zero. Table 6 compares the two assumptions at $\alpha = 0.01$, a value chosen for illustration and not estimated. Empirical estimation of α from county-level default data across drought and non-drought years is identified as a pilot deliverable in Section 8.

Table 6: Tail comparison under Poisson and negative binomial assumptions

Pool size (n)	**λ**	**Poisson 99th pct.**	**Negative binomial 99th pct. (α = 0.01)**	**Tail inflation**
500	58.07	76	80	+5.3%
1,000	116.13	141	153	+8.5%
2,000	232.26	268	297	+10.8%

Tail inflation grows with pool size because the overdispersion term $\alpha\lambda^2$ grows quadratically in n while the Poisson term grows linearly. Larger pools do not diversify away a correlated climate shock. Reserve figures in Section 7.2 are computed on the Poisson basis for comparability with the base case. The negative binomial figures should be read as the stress case, and a prudent implementation would reserve to the higher of the two.

That last point deserves emphasis because it runs against the usual intuition. Pooling diversifies idiosyncratic risk, and the Poisson model reflects that: the coefficient of variation falls as the pool grows. It does not diversify a common shock. Under the negative binomial the coefficient of variation converges to $\sqrt{\alpha}$ rather than to zero, so a pool covering drought-correlated defaults faces an irreducible floor on relative volatility no matter how large it becomes. This is the analytical reason why reinsurance is treated as structural in Section 7.2 rather than as a refinement.

6.4 Risk-tiered credibility scoring

Uniform premium pricing creates adverse selection. Higher-risk borrowers find the product attractive at a price calibrated for the average, while lower-risk borrowers opt out, progressively deteriorating the pool’s risk profile. ArdhiGuard addresses this through risk-tiered pricing, where each borrower’s observed credit characteristics determine a rating factor applied to the base gross premium.

6.4.1 Rating factor derivation

The credibility-adjusted premium for borrower i is:

$$P_i = GP \times C_i \tag{9}$$

where GP is the base gross premium from Equation (5) and C_i is the rating factor for borrower i, derived from the observed default rates by credit score category in Table 3:

$$C_i = \frac{q_i}{q} \tag{10}$$

where q_i is the observed default rate for borrower i's credit score category and q = 0.11613 is the pool average. This is a multiplicative rating factor model and ensures that each tier's premium is actuarially consistent with its observed claim rate. Because the rating factors are constructed to average to unity under the observed class distribution, the tiering is revenue-neutral at the pool level: it redistributes the premium across borrowers rather than raising it.

6.4.2 Risk-tiered premium schedule

Table 7: Risk-tiered premium schedule by credit score category

Credit score	**Default rate (q_i)**	**Rating factor (C_i)**	**Annual premium (P_i)**
Excellent	10.17%	0.8758	9,342.31
Good	10.63%	0.9154	9,764.73
Fair	11.42%	0.9834	10,490.09
Poor	12.48%	1.0747	11,464.01

All premiums in monetary units (MU). See Section 5.1.

The spread from 9,342.31 for Excellent to 11,464.01 for Poor represents a 22.7 per cent differential, which is meaningful for risk segmentation while remaining within a range that does not price the highest-risk tier out of the pool. That balance is the design constraint: segmentation that is too aggressive defeats the social purpose of the product by excluding exactly the borrowers most likely to lose land.

6.4.3 Loan purpose as a supplementary factor

Table 4 shows that default rates also vary by loan purpose, from 10.23 per cent for home loans to 12.33 per cent for business loans. An analogous set of loan-purpose rating factors can be derived using the same methodology. In full implementation both credit score and loan purpose factors would be combined in a two-way multiplicative rating structure, with the caveat that the two variables are unlikely to be independent and a generalised linear model would be preferred to a naive product of one-way factors. This extension is recommended for the pilot phase and is beyond the scope of the present modelling.

6.5 Dynamic SPI-linked premium repricing

The dynamic climate repricing mechanism is the principal methodological contribution of the ArdhiGuard framework. Unlike conventional credit insurance products, which maintain fixed premiums throughout the policy period, ArdhiGuard adjusts premiums in response to changing drought conditions using the Standardised Precipitation Index.

6.5.1 The Standardised Precipitation Index

The SPI is a widely accepted meteorological indicator measuring the deviation of observed precipitation from the long-term average for a given location and time window, expressed in standard deviations (World Meteorological Organisation, 2012). The WMO classifies drought severity as normal conditions where SPI is at or above −1.0, moderate drought where SPI lies between −1.5 and −1.0, severe drought where SPI lies between −2.0 and −1.5, and extreme drought where SPI falls below −2.0.

Kenya's arid and semi-arid lands, which cover approximately 80 per cent of the country and support over 70 per cent of the national livestock herd, are particularly susceptible to SPI excursions below −1.5. The drought cycles of 2008 to 2011, 2016 to 2017 and 2020 to 2023 each saw county-level SPI values below −2.0 sustained over multiple consecutive months (Kenya Meteorological Department, 2022).

6.5.2 The Climate Adjustment Factor

The Climate Adjustment Factor translates drought severity into a premium multiplier. It is defined as a capped piecewise linear function of the SPI:

$$\mathrm{CAF} = 1, \ \text{if SPI} \geq -1.0 \tag{11}$$

$$CAF = min[1 + \beta(-SPI - 1), CAF_{\max} =], \ if\ SPI < -1.0 \tag{12}$$

where β is the climate sensitivity coefficient and $\mathrm{CAF}_{\max}$ is a regulatory and affordability ceiling. The function is continuous at the boundary: at SPI = −1.0 both branches yield CAF = 1. For SPI below −1.0, each additional unit of drought severity increases the premium by β × 100 per cent until the cap binds. The climate-adjusted gross premium is:

$$GP_{new} = GP_{base} \times CAF \quad (13)$$

6.5.3 The cap and why it is necessary

An uncapped multiplier is neither approvable nor coherent. It is not approvable because the IRA would be unlikely to authorise an unbounded premium adjustment mechanism under the Microinsurance Regulations. It is not coherent because a borrower already experiencing extreme drought is, by construction, least able to absorb a premium increase, so an uncapped CAF would price cover away at exactly the moment it is needed. A ceiling of $CAF_{max} = 1.30$ is adopted here. At β = 0.20 this binds at SPI = −2.5, beyond the WMO extreme drought threshold. Losses arising beyond that point are transferred to reinsurance rather than to the borrower, which is consistent with the reinsurance requirement established in Section 7.2.

6.5.4 The climate sensitivity coefficient

The coefficient β quantifies the marginal impact of a one-unit increase in drought severity on the required premium. In the current model β = 0.20 is a calibration assumption, implying a 20 per cent premium increase per unit of drought severity below the threshold. This value is illustrative pending empirical calibration.

Calibration route for β. *The appropriate value of β should be estimated by regressing the historical excess default rate during drought periods on the contemporaneous SPI value, using county-level data from the Kenya Meteorological Department (2022) matched against loan default records from a partner institution. The regression should include county fixed effects to absorb time-invariant differences in agro-ecological exposure, and should test for asymmetry between drought and flood conditions, since the SPI is symmetric but the underlying agricultural damage function is not. This calibration is identified as a priority deliverable for the pilot phase in Section 8.*

6.5.5 Worked example and full schedule

Assume a county experiences an SPI of −1.5, the boundary of moderate drought, during a policy year, and the base gross premium is 10,667.17 MU. With β = 0.20:

$$CAF = 1 + 0.20 \times (1.5 - 1) = 1.10 (14)$$

$$GP_{new} = 10{,}667.17 \times 1.10 = 11{,}733.89\, MU \quad (15)$$

The premium rises 10 per cent in response to moderate drought conditions. Table 8 presents climate-adjusted premiums across the full WMO drought severity scale.

Table 8: Climate-adjusted premiums at each drought severity level ($\beta = 0.20$, $CAF_{max} = 1.30$)

WMO classification	SPI applied	CAF	Adjusted GP (MU)	Change
Normal	≥ −1.0	1.000	10,667.17	Baseline
Moderate drought	−1.5	1.100	11,733.89	+10.0%
Severe drought	−2.0	1.200	12,800.60	+20.0%
Extreme drought	≤ −2.5	1.300	13,867.32	+30.0%

The lower bound of each WMO range is applied rather than the midpoint, which is the conservative choice. The extreme drought row is evaluated at the cap, which binds at SPI = −2.5.

When climatic conditions improve and the SPI returns above −1.0, the CAF reverts to 1 and premiums return to the base level. This symmetric adjustment preserves actuarial fairness: borrowers pay more during periods of elevated climate risk and less when risk subsides. Symmetry is also what distinguishes the mechanism from a surcharge. A one-way ratchet would be a price increase dressed as a risk adjustment.

6.5.6 Repricing frequency

The SPI is recalculated by the Kenya Meteorological Department monthly for each county. ArdhiGuard's adjustment mechanism is therefore designed to reprice at policy renewal, using the 3-month SPI observed over the preceding quarter, which is the standard agricultural drought indicator. Mid-term repricing, where a drought materialises after policy inception, would require a product endorsement mechanism under IRA microinsurance guidelines and is a regulatory consideration for the implementation phase.

6.6 Sensitivity analysis

To evaluate the robustness of the pricing model, a sensitivity analysis examines how changes in the default probability, recovery rate and climatic conditions affect the computed premium. Three scenarios are evaluated, with all figures derived from Equations (4), (5), (12) and (13).

Table 9: Sensitivity analysis, three pricing scenarios

Scenario	q	r	CAF	EL	GP pre-CAF	GP final
Optimistic	0.08000	55%	1.00	4,592.84	5,511.41	5,511.41
Base case	0.11613	40%	1.00	8,889.31	10,667.17	10,667.17
Conservative	0.15000	25%	1.20	14,352.63	17,223.16	20,667.79

All figures in monetary units. $L = 127{,}578.9$ and $\theta = 0.20$ in all scenarios. The conservative scenario applies CAF = 1.20, corresponding to SPI = −2.0.

The premium range spans from 5,511.41 under the optimistic scenario to 20,667.79 under the conservative one, a range of close to four-fold. This reflects the genuine climate-credit risk faced by land-secured borrowers in drought-prone Kenyan counties. In the conservative scenario the combined effect of a higher default probability, lower recovery and climate adjustment produces a premium approximately twice the base case, which underscores the role of the CAF mechanism in aligning premiums with actual risk exposure. It also illustrates the affordability constraint directly: a near four-fold price range is not something a low-income borrower can absorb, which is why the cap in Section 6.5.3 and the reinsurance arrangement in Section 7.2 are load-bearing parts of the design rather than afterthoughts.

6.7 Model implementation framework

The ArdhiGuard pricing process follows five sequential steps.

Step 1, data collection. Borrower-level data on loan amount, credit score, income, debt-to-income ratio and loan purpose are collected at enrolment.

Step 2, parameter estimation. Pool-level q, average exposure L, recovery rate r and loading θ are estimated or assumed.

Step 3, base premium. $GP = q \times L \times (1 - r) \times (1 + \theta)$.

Step 4, credibility adjustment. $P_i = GP \times C_i$ is applied to produce the borrower-specific premium.

Step 5, climate adjustment. At renewal, the county SPI is observed and $GP_{final} = P_i \times CAF$ is applied.

7. FINANCIAL VIABILITY

7.1 Revenue streams

ArdhiGuard generates revenue through three complementary streams.

Insurance premiums, approximately 65 per cent of revenue. Annual premiums from the borrower pool, tiered by credit risk and collected in twelve monthly mobile money instalments. This is the primary recurring stream.

Risk assessment fees, approximately 20 per cent of revenue. Banks and microfinance institutions pay a per-loan fee for risk scores at loan origination, creating a business-to-business stream that does not depend on borrower enrolment volumes.

Data and analytics, approximately 15 per cent of revenue. Anonymised, aggregated portfolio risk reports for institutional investors, development finance institutions and government agencies.

These proportions are planning assumptions and are not derived from the model.

7.2 Fund solvency and reserve requirements

The protection fund must satisfy two conditions. It must be sufficient to pay claims with a defined probability, and it must meet the IRA's minimum solvency margin of 150 per cent under the Insurance Act, Cap. 487. The required reserve is:

$$\text{Required reserve} = N_{99} \times L(1 - r) \times s \qquad (16)$$

where N_{99} is the 99th percentile of the claim count distribution, $L(1 - r) = 76{,}547.34$ MU is the average net loss per claim, and $s = 1.5$ is the safety multiplier for IRA compliance. Table 10 sets the resulting requirement against annual premium income at each modelled pool size.

Table 10: Required reserve and premium coverage at illustrative pool sizes

Pool size (n)	N_{99}	Required reserve (MU)	Annual premium income (MU)	Coverage ratio
500	76	8,726,397	5,333,585	0.61
1,000	141	16,189,762	10,667,170	0.66
2,000	268	30,772,031	21,334,340	0.69

Annual premium income computed as n × GP at the base premium of 10,667.17 MU. Coverage ratio is annual premium income divided by required reserve. Poisson basis; the negative binomial figures in Table 6 would raise the requirement further.

7.2.1 A structural finding: the reserve gap does not close with scale

The coverage ratio improves as the pool grows, from 0.61 at 500 members to 0.69 at 2,000, because the 99th percentile grows more slowly than the mean. It does not, however, converge to one. As n becomes large, N_{99} approaches $\lambda = nq$, so the required reserve approaches $n \times q \times L(1 - r) \times s = n \times EL \times 1.5$, while premium income is $n \times EL \times 1.2$. The coverage ratio therefore has an upper limit of $1.2 / 1.5 = 0.80$, which is independent of pool size.

What this means. *The reserve requirement cannot be met out of a single year's premium income at any scale. This is not a defect of the parameterisation; it follows arithmetically from setting a loading of 20 per cent against a solvency multiplier of 150 per cent. The gap must be closed from somewhere other than current premium. Retained surplus is the obvious candidate, but on the loading decomposition in Section 6.1.4 only the 5 per cent contingency component accretes to reserves, which implies a build period of roughly thirty years from a standing start. That is not a viable route. The conclusion is that seed capital and quota share reinsurance are structural requirements of the design, not optional refinements, and the earlier suggestion that the pool becomes self-sustaining at 2,000 to 3,000 members does not follow from the model and is withdrawn.*

Three levers can close the gap, and they trade off against each other. Raising θ improves the ratio but works directly against affordability, which is the product's reason for existing. Quota share reinsurance reduces the retained N_{99} and is the lever most consistent with the design, since it also addresses the correlated-shock problem identified in Section 6.3.1 that no amount of pooling resolves. Seed capital bridges the early years but does not change the steady state. The realistic configuration is a modest loading increase combined with reinsurance, with seed capital covering the first three to five years. A founding contribution in the order of KES 10 million has been indicated by prospective partners, though this figure cannot be reconciled against Table 10 until the model is recalibrated in Kenya Shillings, for the reasons set out in Section 5.1.

8. SOCIAL IMPACT

ArdhiGuard is not merely a commercial insurance product. It is a poverty prevention mechanism operating at the intersection of financial services, agricultural resilience and social protection. Table 11 sets out the estimated impact of a 5,000-member pool at scale.

Table 11: Estimated social impact, 5,000-member pool

Impact dimension	**Estimated annual outcome**	**Basis**
Borrowers entering default	581	$5{,}000 \times q = 5{,}000 \times 0.11613$
Families protected from land loss	349	581 × 60% assumed intervention success rate
Children kept in school	698	2 children per household (KNBS)
Agricultural land preserved	1,745 acres	5 acres per household
Food production protected	8,725 tonnes	5 tonnes per acre per year
Loan value protected	44.5 million MU	$349 \times L = 349 \times 127{,}578.9$

All figures are indicative and derive from the model's own parameters combined with KNBS household and agricultural statistics (Kenya National Bureau of Statistics, 2023). The 60 per cent intervention success rate is an assumption, not an estimate, and is the single largest source of uncertainty in this table. Loan value is stated in monetary units for the reasons given in Section 5.1.

Beyond these quantified outcomes, ArdhiGuard addresses a structural inequity in Kenya's financial system: the asymmetry of risk between lenders and borrowers in land-secured credit. Under the current system the lender holds first-priority security and the borrower absorbs all downside risk. ArdhiGuard rebalances this asymmetry by introducing a third party, the insurance pool, that shares the borrower's risk without requiring the lender to relinquish their security position. All three parties benefit. The borrower keeps the land, the lender recovers the loan and the community retains a productive agricultural unit.

9. LIMITATIONS AND DIRECTIONS FOR FURTHER RESEARCH

Seven limitations define the research agenda for the pilot phase. They are stated in descending order of materiality.

9.1 Dataset provenance

The empirical dataset is denominated in unspecified monetary units and drawn from a non-Kenyan lending market. All premium figures in this paper are methodologically illustrative rather than directly applicable to Kenyan borrowers. Recalibration using KES-denominated data from a domestic microfinance institution or SACCO is the highest-priority next step and conditions every other quantitative claim made here.

9.2 The risk engine weights are not estimated

The six weights in Table 1, the 15 per cent intervention threshold and the 60 to 90 day lead time are expert judgement rather than fitted quantities. The paper therefore specifies what the engine measures but makes no validated claim about how well it predicts. Fitting a penalised logistic model to the available records, reporting standardised coefficients and validating out-of-sample would convert this layer from a specification into a result, and is the change that would most strengthen the framework.

9.3 Recovery rate assumption

The recovery rate of 40 per cent is assumed from general credit risk literature relating to partially secured retail exposures, which is not a like-for-like benchmark for land-secured lending. As set out in Section 6.1.3, two biases operate in opposite directions and the net direction cannot be signed without domestic data. Land Registry auction records should be used to estimate this parameter empirically.

9.4 Climate sensitivity coefficient

The value $\beta = 0.20$ is a calibration assumption. Empirical estimation from matched Kenya Meteorological Department drought data and partner institution default records is necessary before the CAF mechanism can be regarded as quantitatively grounded rather than structurally motivated.

9.5 Correlated tail risk and reinsurance

Section 6.3.1 shows that a correlated climate shock is not diversified away by pool growth, and Section 7.2 shows that the reserve requirement cannot be met from premium income at any scale. A severe, prolonged drought spanning multiple counties simultaneously could produce claim frequencies well beyond the 99th percentile modelled here. A quota share reinsurance treaty and a government backstop facility, analogous to the Agriculture Insurance Fund, are proposed as the mitigation route. Specifying the treaty structure and pricing the retained layer is outstanding work.

9.6 Rating structure

The rating model uses credit score as the sole differentiating variable. Loan purpose shows a comparable range of default rates and should be incorporated as a second dimension, preferably through a generalised linear model rather than a product of one-way factors, since the two variables are unlikely to be independent.

9.7 Behavioural response and moral hazard

The model treats the default probability q as exogenous to the existence of the product. This is unlikely to hold. A borrower who knows a protection pool stands behind them may sustain repayment effort differently, in either direction, and the early warning contact itself is an intervention whose effect on q is unmeasured. The pilot should be designed so that this effect is identifiable rather than confounded with selection into the pool.

10. CONCLUSION

The intersection of predictive analytics, actuarial science and mobile money infrastructure is not a future possibility. It is a present necessity in markets where financial exclusion, climate risk and asset vulnerability converge. Kenya's land foreclosure crisis is precisely this intersection in its most human form: a quantifiable, predictable risk with catastrophic and irreversible consequences for those who bear it.

ArdhiGuard demonstrates that actuarial science has the tools to address this crisis. The expected loss framework, the Poisson claim frequency model with its negative binomial extension, the multiplicative credibility rating structure and the dynamic climate adjustment mechanism together constitute a financially rigorous and empirically grounded specification. The early warning layer adds a dimension of proactive intervention that conventional insurance does not provide. The mobile money delivery platform ensures accessibility at the income levels where the risk is most acute.

This paper contributes three specific results. First, an empirically grounded expected loss pricing model for land-secured credit insurance, calibrated to 255,347 borrower records and producing a base annual gross premium of 10,667.17 monetary units with a 22.7 per cent risk-tiered spread.

Second, a dynamic Climate Adjustment Factor that links the Standardised Precipitation Index to insurance premiums, with an explicit cap justified on both regulatory and affordability grounds, enabling actuarially fair repricing as drought conditions evolve. Third, a structural solvency result showing that under a 20 per cent loading and a 150 per cent solvency margin, premium income converges to 80 per cent of the required reserve regardless of pool size, so that seed capital and quota share reinsurance are structural requirements of the design rather than optional refinements.

The third result is the least comfortable of the three and, for that reason, the most useful. A framework that only reported the premium schedule would leave a reader with the impression that the pool funds itself once it is large enough. It does not, and the arithmetic that shows why also points directly at the remedy. The correlated nature of drought risk means that pooling alone cannot deliver solvency, and risk transfer is required. Naming that early is the difference between a design that survives regulatory review and one that does not.

The theme of this year's convention, Actuaries Beyond Boundaries, captures what ArdhiGuard represents. Land protection is not a traditional actuarial product. Default prediction is not a traditional actuarial role. Poverty prevention is not a traditional actuarial outcome. ArdhiGuard crosses all three boundaries deliberately, on the argument that the future of the actuarial profession in Africa lies in exactly this kind of crossing: applying the rigour of the discipline where it is needed most.

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APPENDIX A: SUMMARY OF MODEL PARAMETERS

Table A1: Summary of model parameters

Symbol	Parameter	Value	Status
q	Default probability	0.11613	Estimated from data
L	Exposure at default	127,578.9 MU	Estimated from data
r	Recovery rate	0.40	Assumed; see 6.1.3
θ	Loading factor	0.20	Assumed; see 6.1.4
EL	Expected loss	8,889.31 MU	Derived
GP	Base annual gross premium	10,667.17 MU	Derived
C_i	Rating factors	0.8758 to 1.0747	Derived from data
α	Overdispersion parameter	0.01	Illustrative; to be estimated
β	Climate sensitivity coefficient	0.20	Assumed; to be calibrated
CAF_{max}	Climate adjustment cap	1.30	Design choice; see 6.5.3
s	Solvency safety multiplier	1.5	IRA requirement

APPENDIX B:R OUTPUT